



CHAPTE 4

QUADRATIC EQUATIONS

Quadratic Equation:

A quadratic equation in one variable x is an equation of the form

$$ax^2 + bx + c = 0 \text{ where } a, b \text{ and } c \text{ are real numbers and } a \neq 0.$$

Example:

$$4x^2 + 5x + 3 = 0, \quad x^2 - 7x + 10 = 0 \text{ are quadratic equations.}$$

Note:

- (i) $ax^2 + bx + c = 0$ is called **standard form** of quadratic equation.
- (ii) Another name for quadratic equation is **Second Degree Polynomial in x** .

Solution of Quadratic Equation:

There are three techniques for solving a quadratic equation.

(i) By Factorization:

In this method we factorize the polynomial $ax^2 + bx + c$ into two linear factors.

We use the fact that, if $ab = 0$ then either $a = 0$ or $b = 0$

(ii) By Completing Square:

In this method we form, by construction, the perfect square of a linear term involving variables. Then solution is found by extracting square roots.

This method is used, when quadratic equation is not easily factorable.

(iii) By Quadratic Formula:

In this method we can find the solution by applying a formula known as “Quadratic Formula”

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Every quadratic equation can be solved by using quadratic formula.

Derivation of Quadratic Formula:

For given quadratic equation $ax^2 + bx + c = 0$; $a \neq 0$

Step#1:

Divide both sides of equation by a

$$x^2 + \frac{b}{a}x + \frac{c}{a} = 0$$

Step#2:

Shift constant term to right side of equation

$$x^2 + \frac{b}{a}x = -\frac{c}{a}$$

Step#3:

Add $\left(\frac{b}{2a}\right)^2$ (the square of half of the coefficient of x) to both sides.

$$x^2 + \frac{b}{a}x + \left(\frac{b}{2a}\right)^2 = -\frac{c}{a} + \left(\frac{b}{2a}\right)^2$$

$$(x)^2 + 2(x)\left(\frac{b}{2a}\right) + \left(\frac{b}{2a}\right)^2 = \frac{b^2}{4a^2} - \frac{c}{a}$$

$$\left(x + \frac{b}{2a}\right)^2 = \frac{b^2 - 4ac}{4a^2}$$

$$x + \frac{b}{2a} = \pm \frac{\sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-b}{2a} \pm \frac{\sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Quadratic formula:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Note: The solutions of an equation are also called its roots.

Exercise 4.1**Solve the following equations by Factorization:**

Q.1 $3x^2 + 4x + 1 = 0$

Solution:

$$3x^2 + 4x + 1 = 0$$

$$3x^2 + 3x + x + 1 = 0$$

$$3x(x+1) + 1(x+1) = 0$$

$$(x+1)(3x+1) = 0$$

Either	$x+1=0$	or	$3x+1=0$
	$x=-1$	;	$3x=-1$
		;	$x=-\frac{1}{3}$

$$\therefore \text{Solution Set} = \left\{ -\frac{1}{3}, -1 \right\}$$

Q.2 $x^2 + 7x + 12 = 0$

Solution:

$$x^2 + 7x + 12 = 0$$

$$x^2 + 4x + 3x + 12 = 0$$

$$x(x+4) + 3(x+4) = 0$$

$$(x+4)(x+3) = 0$$

Either	$x+4=0$	or	$x+3=0$
	$x=-4$	;	$x=-3$

$$\therefore \text{Solution Set} = \{-4, -3\}$$

Q.3 $9x^2 - 12x - 5 = 0$

Solution:

$$9x^2 - 12x - 5 = 0$$

$$9x^2 - 15x + 3x - 5 = 0$$

$$3x(3x-5) + 1(3x-5) = 0$$

$$(3x-5)(3x+1) = 0$$

Either	$3x-5=0$	or	$3x+1=0$
	$3x=5$	;	$3x=-1$

$$x = \frac{5}{3} \quad ; \quad x = \frac{-1}{3}$$

$$\therefore \text{Solution Set is } \left\{ \frac{-1}{3}, \frac{5}{3} \right\}$$

Q.4 $x^2 - x = 2$

Solution:

$$x^2 - x = 2$$

$$x^2 - x - 2 = 0$$

$$x^2 - 2x + x - 2 = 0$$

$$x(x-2) + 1(x-2) = 0$$

$$(x-2)(x+1) = 0$$

$$\begin{array}{lll} \text{Either} & x-2=0 & \text{or} \quad x+1=0 \\ & x=2 & ; \quad x=-1 \end{array}$$

$$\therefore \text{Solution Set} = \{-1, 2\}$$

Q.5 $x(x+7) = (2x-1)(x+4)$

Solution:

$$x(x+7) = (2x-1)(x+4)$$

$$x^2 + 7x = 2x^2 + 8x - x - 4$$

$$2x^2 + \cancel{8x} - \cancel{x} - 4 - x^2 - \cancel{7x} = 0$$

$$x^2 - 4 = 0$$

$$(x)^2 - (2)^2 = 0$$

$$(x-2)(x+2) = 0$$

$$\begin{array}{lll} \text{Either} & x-2=0 & \text{or} \quad x+2=0 \\ & x=2 & ; \quad x=-2 \end{array}$$

$$\therefore \text{Solution Set} = \{-2, 2\}$$

Q.6 $\frac{x}{x+1} + \frac{x+1}{x} = \frac{5}{2}, x \neq -1, 0$

Solution:

$$\frac{x}{x+1} + \frac{x+1}{x} = \frac{5}{2}$$

Multiplying both sides by ' $2x(x+1)$ '

$$2x^2 + 2(x+1)^2 = 5x(x+1)$$

$$2x^2 + 2(x^2 + 1 + 2x) = 5x^2 + 5x$$

$$2x^2 + 2x^2 + 2 + 4x = 5x^2 + 5x$$

$$5x^2 + 5x - 2x^2 - 2x^2 - 4x - 2 = 0$$

$$x^2 + x - 2 = 0$$

$$x^2 + 2x - x - 2 = 0$$

$$x(x+2) - 1(x+2) = 0$$

$$(x+2)(x-1) = 0$$

$$\text{Either } \begin{array}{l} x+2=0 \\ x=-2 \end{array} \quad \text{or} \quad \begin{array}{l} x-1=0 \\ x=1 \end{array}$$

$$\therefore \text{ Solution Set} = \{-2, 1\}$$

Q.7 $\frac{1}{x+1} + \frac{2}{x+2} = \frac{7}{x+5}, x \neq -1, -2, -5$

Solution:

$$\frac{1}{x+1} + \frac{2}{x+2} = \frac{7}{x+5}$$

Multiplying Both sides by $(x+1)(x+2)(x+5)$

$$(x+2)(x+5) + 2(x+1)(x+5) = 7(x+1)(x+2)$$

$$x^2 + 7x + 10 + 2(x^2 + 6x + 5) = 7(x^2 + 3x + 2)$$

$$x^2 + 7x + 10 + 2x^2 + 12x + 10 = 7x^2 + 21x + 14$$

$$7x^2 - 3x^2 + 21x - 19x + 14 - 20 = 0$$

$$4x^2 + 2x - 6 = 0$$

$$2x^2 + x - 3 = 0$$

Q Dividing both sides by 2

$$2x^2 + 3x - 2x - 3 = 0$$

$$x(2x+3) - 1(2x+3) = 0$$

$$(2x+3)(x-1) = 0$$

$$\text{Either } \begin{array}{l} 2x+3=0 \\ x = \frac{-3}{2} \end{array} \quad \text{or} \quad \begin{array}{l} x-1=0 \\ x=1 \end{array}$$

$$\therefore \text{ Solution Set} = \left\{ \frac{-3}{2}, 1 \right\}$$

Q.8 $\frac{a}{ax-1} + \frac{b}{bx-1} = a+b, x \neq \frac{1}{a}, \frac{1}{b}$

Solution:

$$\frac{a}{ax-1} + \frac{b}{bx-1} = a+b$$

$$\frac{a}{ax-1} - b + \frac{b}{bx-1} - a = 0$$

$$\frac{a-b(ax-1)}{ax-1} + \frac{b-a(bx-1)}{bx-1} = 0$$

$$\frac{a-abx+b}{ax-1} + \frac{b-abx+a}{bx-1} = 0$$

$$\frac{(bx-1)[a-abx+b] + (ax-1)[b-abx+a]}{(ax-1)(bx-1)} = 0$$

$$\frac{(a+b-abx)[bx-1+ax-1]}{(ax-1)(bx-1)} = 0$$

$$(a+b-abx)(bx+ax-2) = 0$$

Either $a+b-abx=0$ or $bx+ax-2=0$

$$abx = a+b ; x(a+b) = 2$$

$$x = \frac{a+b}{ab} ; x = \frac{2}{a+b}$$

$$\therefore \text{Solution set} = \left\{ \frac{a+b}{ab}, \frac{2}{a+b} \right\}$$

Solve the following Equations by completing square:

Q.9 $x^2 - 2x - 899 = 0$

Solution:

$$x^2 - 2x - 899 = 0$$

$$x^2 - 2x = 899$$

Adding 1^2 on both sides

$$x^2 - 2x + 1^2 = 899 + 1^2$$

$$(x-1)^2 = 900$$

Taking square root on both sides

$$x-1 = \pm 30$$

Either $x-1=30$ or $x-1=-30$

$$x=31 ; x=-29$$

$$\therefore \text{Solution Set} = \{-29, 31\}$$

Q.10 $x^2 + 4x - 1085 = 0$

Solution:

$$x^2 + 4x - 1085 = 0$$

$$x^2 + 4x = 1085$$

Adding $(2)^2$ on both sides

$$x^2 + 4x + (2)^2 = 1085 + (2)^2$$

$$(x + 2)^2 = 1089$$

Taking square root on both sides

$$x + 2 = \pm 33$$

$$\begin{array}{llll} \text{Either} & x + 2 = 33 & \text{or} & x + 2 = -33 \\ & x = 31 & ; & x = -35 \end{array}$$

$$\therefore \text{Solution Set} = \{-35, 31\}$$

Q.11 $x^2 + 6x - 567 = 0$

Solution:

$$x^2 + 6x - 567 = 0$$

$$x^2 + 6x = 567$$

Adding $(3)^2$ on both sides

$$x^2 + 6x + (3)^2 = 567 + (3)^2$$

$$(x + 3)^2 = 576$$

Taking square root on both sides

$$(x + 3) = \pm 24$$

$$\begin{array}{llll} \text{Either} & x + 3 = 24 & \text{or} & x + 3 = -24 \\ & x = 21 & ; & x = -27 \end{array}$$

$$\therefore \text{Solution Set} = \{-27, 21\}$$

Q.12 $x^2 - 3x - 648 = 0$

Solution:

$$x^2 - 3x - 648 = 0$$

$$x^2 - 3x = 648$$

Adding $\left(\frac{3}{2}\right)^2$ on both sides

$$x^2 - 3x + \left(\frac{3}{2}\right)^2 = 648 + \left(\frac{3}{2}\right)^2$$

$$\left(x - \frac{3}{2}\right)^2 = 648 + \frac{9}{4}$$

$$\left(x - \frac{3}{2}\right)^2 = \frac{2592 + 9}{4}$$

$$\left(x - \frac{3}{2}\right)^2 = \frac{2601}{4}$$

Taking square root on both sides

$$x - \frac{3}{2} = \pm \frac{51}{2}$$

$$\begin{array}{ll} \text{Either} & x - \frac{3}{2} = \frac{51}{2} \quad \text{or} \quad x - \frac{3}{2} = \frac{-51}{2} \\ & x = \frac{51}{2} + \frac{3}{2} = \frac{54}{2} \quad ; \quad x = \frac{-51}{2} + \frac{3}{2} = -\frac{48}{2} \\ & x = 27 \quad ; \quad x = -24 \end{array}$$

$$\therefore \text{Solution Set} = \{-24, 27\}$$

Q.13 $x^2 - x - 1806 = 0$

Solution:

$$x^2 - x - 1806 = 0$$

$$x^2 - x = 1806$$

Adding $\left(\frac{1}{2}\right)^2$ on both sides

$$x^2 - x + \left(\frac{1}{2}\right)^2 = 1806 + \left(\frac{1}{2}\right)^2$$

$$\left(x - \frac{1}{2}\right)^2 = \frac{7224 + 1}{4}$$

$$\left(x - \frac{1}{2}\right)^2 = \frac{7225}{4}$$

$$x - \frac{1}{2} = \pm \frac{85}{2}$$

$$\begin{array}{ll} \text{Either} & x - \frac{1}{2} = \frac{85}{2} \quad \text{or} \quad x - \frac{1}{2} = \frac{-85}{2} \\ & x = \frac{85}{2} + \frac{1}{2} = \frac{86}{2} \quad ; \quad x = \frac{-85}{2} + \frac{1}{2} = -\frac{84}{2} \\ & x = 43 \quad ; \quad x = -42 \end{array}$$

$$\therefore \text{Solution Set is} = \{-42, 43\}$$

Q.14 $2x^2 + 12x - 110 = 0$

Solution:

$$2x^2 + 12x - 110 = 0$$

Dividing both sides by 2

$$x^2 + 6x - 55 = 0$$

$$x^2 + 6x = 55$$

Adding ' $(3)^2$ ', on both sides

$$x^2 + 6x + (3)^2 = 55 + (3)^2$$

$$(x+3)^2 = 64$$

Taking square root on both sides

$$x+3 = \pm 8$$

$$\text{Either } \begin{array}{l} x+3=8 \\ x=5 \end{array} \quad \text{or} \quad \begin{array}{l} x+3=-8 \\ x=-11 \end{array}$$

$$\therefore \text{ Solution Set} = \{-11, 5\}$$

Find roots of the following equation by using quadratic formula:

Q.15 $5x^2 - 13x + 6 = 0$

Solution:

$$5x^2 - 13x + 6 = 0$$

By Comparing with $ax^2 + bx + c = 0$, we get,

$$a = 5, \quad b = -13, \quad c = 6$$

By quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-(-13) \pm \sqrt{(-13)^2 - 4(5)(6)}}{2(5)}$$

$$x = \frac{13 \pm \sqrt{169 - 120}}{10}$$

$$x = \frac{13 \pm \sqrt{49}}{10}$$

$$x = \frac{13 \pm 7}{10}$$

$$\text{Either } \begin{array}{l} x = \frac{13+7}{10} = \frac{20}{10} \\ x = 2 \end{array} \quad \text{or} \quad \begin{array}{l} x = \frac{13-7}{10} = \frac{6}{10} \\ x = \frac{3}{5} \end{array}$$

$$\therefore \text{ Solution Set} = \left\{ \frac{3}{5}, 2 \right\}$$

Q.16 $4x^2 + 7x - 1 = 0$

Solution:

$$4x^2 + 7x - 1 = 0$$

By Comparing with $ax^2 + bx + c = 0$, we get,

$$a = 4, b = 7, c = -1$$

By quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-7 \pm \sqrt{(7)^2 - 4(4)(-1)}}{2(4)}$$

$$x = \frac{-7 \pm \sqrt{49 + 16}}{8}$$

$$x = \frac{-7 \pm \sqrt{65}}{8}$$

Either $x = \frac{-7 + \sqrt{65}}{8}$ or $x = \frac{-7 - \sqrt{65}}{8}$

$$\therefore \text{Solution Set} = \left\{ \frac{-7 - \sqrt{65}}{8}, \frac{-7 + \sqrt{65}}{8} \right\}$$

Q.17 $15x^2 + 2ax - a^2 = 0$

Solution:

$$15x^2 + 2ax - a^2 = 0$$

By Comparing with $Ax^2 + Bx + C = 0$, we get,

$$A = 15, B = 2a, C = -a^2$$

By quadratic formula

$$x = \frac{-B \pm \sqrt{B^2 - 4AC}}{2A}$$

$$x = \frac{-2a \pm \sqrt{(2a)^2 - 4(15)(-a^2)}}{2(15)}$$

$$x = \frac{-2a \pm \sqrt{4a^2 + 60a^2}}{30}$$

$$x = \frac{-2a \pm \sqrt{64a^2}}{30}$$

$$x = \frac{-2a \pm 8a}{30}$$

Either $x = \frac{-2a + 8a}{30} = \frac{6a}{30}$ or $x = \frac{-2a - 8a}{30} = -\frac{10a}{30}$

$x = \frac{a}{5}$; $x = \frac{-a}{3}$

$$\therefore \text{Solution Set} = \left\{ \frac{-a}{3}, \frac{a}{5} \right\}$$

Q.18 $16x^2 + 8x + 1 = 0$

Solution:

$$16x^2 + 8x + 1 = 0$$

By Comparing with $ax^2 + bx + c = 0$, we get,

$$a = 16, b = 8, c = 1$$

By quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-8 \pm \sqrt{(8)^2 - 4(16)(1)}}{2(16)}$$

$$x = \frac{-8 \pm \sqrt{64 - 64}}{32}$$

$$x = \frac{-8}{32}$$

$$x = \frac{-1}{4}$$

$$\therefore \text{Solution Set} = \left\{ \frac{-1}{4} \right\}$$

Q.19 $(x-a)(x-b) + (x-b)(x-c) + (x-c)(x-a) = 0$

Solution:

$$(x-a)(x-b) + (x-b)(x-c) + (x-c)(x-a) = 0$$

$$x^2 - ax - bx + ab + x^2 - bx - cx + bc + x^2 - ax - cx + ac = 0$$

$$3x^2 - 2ax - 2bx - 2cx + ab + bc + ca = 0$$

$$3x^2 - 2(a+b+c)x + ab + bc + ca = 0$$

By Comparing with $Ax^2 + Bx + C = 0$, we get,

$$A = 3, B = -2(a+b+c), C = ab + bc + ca$$

By quadratic formula

$$x = \frac{-B \pm \sqrt{B^2 - 4AC}}{2A}$$

$$= \frac{-[-2(a+b+c)] \pm \sqrt{(-2(a+b+c))^2 - 4(3)(ab+bc+ca)}}{2(3)}$$

$$= \frac{2(a+b+c) \pm \sqrt{4[a^2 + b^2 + c^2 + 2ab + 2bc + 2ca - 3ab - 3bc - 3ca]}}{6}$$

$$= \frac{2(a+b+c) \pm 2\sqrt{a^2 + b^2 + c^2 - ab - bc - ca}}{6}$$

$$= \frac{\cancel{2} \left[(a+b+c) \pm \sqrt{a^2+b^2+c^2-ab-bc-ca} \right]}{\cancel{6}^3}$$

$$x = \frac{(a+b+c) \pm \sqrt{a^2+b^2+c^2-ab-bc-ca}}{3}$$

$$\therefore \text{ the Solution Set } = \left\{ \frac{(a+b+c) \pm \sqrt{a^2+b^2+c^2-ab-bc-ca}}{3} \right\}$$

Q.20 $(a+b)x^2 + (a+2b+c)x + b+c = 0$

Solution:

$$(a+b)x^2 + (a+2b+c)x + b+c = 0$$

By Comparing with $Ax^2 + Bx + C = 0$, we get,

$$A = a+b, \quad B = a+2b+c, \quad C = b+c$$

By using quadratic formula

$$x = \frac{-B \pm \sqrt{B^2 - 4AC}}{2A}$$

$$= \frac{-(a+2b+c) \pm \sqrt{(a+2b+c)^2 - 4(a+b)(b+c)}}{2(a+b)}$$

$$= \frac{-(a+2b+c) \pm \sqrt{a^2 + 4b^2 + c^2 + 4ab + 4bc + 2ac - 4(ab + ac + b^2 + bc)}}{2(a+b)}$$

$$= \frac{-(a+2b+c) \pm \sqrt{a^2 + \cancel{4b^2} + c^2 + \cancel{4ab} + \cancel{4bc} + 2ca - \cancel{4ab} - 4ac - \cancel{4b^2} - \cancel{4bc}}}{2(a+b)}$$

$$= \frac{-(a+2b+c) \pm \sqrt{(a-c)^2}}{2(a+b)}$$

$$= \frac{-(a+2b+c) \pm (a-c)}{2(a+b)}$$

Either $x = \frac{-(a+2b+c) + (a-c)}{2(a+b)}$ or $x = \frac{-(a+2b+c) - (a-c)}{2(a+b)}$

$$x = \frac{-2b-2c}{2(a+b)} = \frac{\cancel{2}(b+c)}{\cancel{2}(a+b)} \quad \text{or} \quad x = \frac{-2a-2b}{2(a+b)} = -\frac{\cancel{2}(a+b)}{\cancel{2}(a+b)}$$

$$x = -\frac{b+c}{a+b} \quad \text{or} \quad x = -1$$

$$\therefore \text{ Solution Set } = \left\{ -\frac{b+c}{a+b}, -1 \right\}$$

Solution of Equations Reducible to the Quadratic Equation:-

There are certain types of equations, which do not look to be of degree 2, but they can be reduced to the quadratic form. We shall discuss the solutions of such five types of the equations one by one.

Type-1:

The equations of the form $ax^{2n} + bx^n + c = 0$; $a \neq 0$ can be converted into quadratic equation by substituting $y = x^n$ and $y^2 = x^{2n}$.

Type-2:

The equation of the form $(x+a)(x+b)(x+c)(x+d) = k$, where $a+b=c+d$, substitute

$$x^2 + (a+b)x = y \text{ and } x^2 + (c+d)x = y \text{ (Since } y = x^2 + (a+b)x = x^2 + (c+d)x \text{)}$$

Type-3:**Exponential Equations:**

Equations, in which the variable occurs in exponent, are called exponential equations.

For example.

$$a^{2x} + b.a^x + c = 0, \text{ Substitute } a^x = y$$

Type-4:**Reciprocal equations:**

There are the equations which remain unchanged even when x is replaced by $\frac{1}{x}$.

If equation written in order (ascending or descending) the coefficients of terms equidistant from beginning and end are equal in magnitude. For example

$$ax^2 + bx + c + \frac{b}{x} + \frac{a}{x^2} = 0, \text{ Substitute } x + \frac{1}{x} = y \text{ and } x^2 + \frac{1}{x^2} = y^2 - 2$$

Exercise 4.2

Q.1 $x^4 - 6x^2 + 8 = 0$

Solution:

$$x^4 - 6x^2 + 8 = 0 \quad (i)$$

Put $x^2 = y \Rightarrow x^4 = y^2$

Equation (i) becomes

$$y^2 - 6y + 8 = 0$$

$$y^2 - 4y - 2y + 8 = 0$$

$$y(y - 4) - 2(y - 4) = 0$$

$$(y - 4)(y - 2) = 0$$

Either $y - 4 = 0$ or $y - 2 = 0$

As $y = x^2$; As $y = x^2$

$$x^2 = 4$$
 ; $x^2 = 2$

$$x = \pm 2$$
 ; $x = \pm\sqrt{2}$

$$\therefore \text{Solution Set} = \{-2, -\sqrt{2}, \sqrt{2}, 2\}$$

Q.2 $x^{-2} - 10 = 3x^{-1}$

Solution:

$$x^{-2} - 10 = 3x^{-1} \quad (i)$$

Put $x^{-1} = y \Rightarrow x^{-2} = y^2$

Equation (i) becomes

$$y^2 - 3y - 10 = 0$$

$$y^2 - 5y + 2y - 10 = 0$$

$$y(y - 5) + 2(y - 5) = 0$$

$$(y - 5)(y + 2) = 0$$

Either $y - 5 = 0$ or $y + 2 = 0$

$$y = 5$$
 ; $y = -2$

As $y = x^{-1}$; As $y = x^{-1}$

$$x^{-1} = 5$$
 ; $x^{-1} = -2$

$$x = \frac{1}{5}$$
 ; $x = -\frac{1}{2}$

$$\therefore \text{Solution Set} = \left\{-\frac{1}{2}, \frac{1}{5}\right\}$$

Q.3 $x^6 - 9x^3 + 8 = 0$

Solution:

$$x^6 - 9x^3 + 8 = 0 \quad (i)$$

Put $x^3 = y \Rightarrow x^6 = y^2$

Equation (i) becomes

$$y^2 - 9y + 8 = 0$$

$$y^2 - 8y - y + 8 = 0$$

$$y(y-8) - 1(y-8) = 0$$

$$(y-8)(y-1) = 0$$

Either

$$y-8=0$$

$$y=8$$

As $y = x^3$

$$x^3 = 8 = (2)^3$$

$$(x)^3 - (2)^3 = 0$$

$$(x-2)(x^2 + 2x + 4) = 0$$

Either

$$x-2=0$$

$$\boxed{x=2}$$

or $x^2 + 2x + 4 = 0$;

or $a=1, b=2, c=4$;

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-2 \pm \sqrt{(2)^2 - 4(1)(4)}}{2(1)}$$

$$x = \frac{-2 \pm \sqrt{4-16}}{2}$$

$$x = \frac{-2 \pm \sqrt{-12}}{2}$$

$$x = \frac{-2 \pm \sqrt{4 \times (-3)}}{2}$$

$$x = \frac{-2 \pm 2\sqrt{-3}}{2}$$

$$x = \frac{\cancel{2}(-1 \pm \sqrt{-3})}{\cancel{2}}$$

$$\boxed{x = -1 \pm \sqrt{3}i}$$

or

$$y-1=0$$

;

$$y=1$$

;

As $y = x^3$

;

$$x^3 = 1 = (1)^3$$

;

$$(x)^3 - (1)^3 = 0$$

;

$$(x-1)(x^2 + x + 1) = 0$$

Either

$$x-1=0$$

$$\boxed{x=1}$$

or

$$x^2 + x + 1 = 0$$

or

$$a=1, b=1, c=1$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-1 \pm \sqrt{(1)^2 - 4(1)(1)}}{2(1)}$$

$$x = \frac{-1 \pm \sqrt{-3}}{2}$$

$$\boxed{x = \frac{-1 \pm \sqrt{3}i}{2}}$$

$$\therefore \text{Solution Set} = \left\{ 1, 2, -1 \pm \sqrt{3}i, \frac{-1 \pm \sqrt{3}i}{2} \right\}$$

Q.4 $8x^6 - 19x^3 - 27 = 0$

Solution:

$$8x^6 - 19x^3 - 27 = 0 \quad (i)$$

Put $x^3 = y \Rightarrow x^6 = y^2$

Equation (i) becomes

$$8y^2 - 19y - 27 = 0$$

$$8y^2 - 27y + 8y - 27 = 0$$

$$y(8y - 27) + 1(8y - 27) = 0$$

$$(8y - 27)(y + 1) = 0$$

Either

$$8y - 27 = 0$$

or $y + 1 = 0$

$$8y = 27$$

; $y + 1 = 0$

$$y = \frac{27}{8}$$

; $y = -1$

As $y = x^3$

; As $y = x^3$

$$x^3 = \frac{27}{8} = \left(\frac{3}{2}\right)^3$$

; $x^3 = -1 \Rightarrow x^3 + 1 = 0 \Rightarrow (x)^3 + (1)^3 = 0$

$$(x)^3 - \left(\frac{3}{2}\right)^3 = 0$$

; $(x+1)(x^2 - x + 1) = 0$

; $x+1=0$ or $x^2 - x + 1 = 0$

$$\left(x - \frac{3}{2}\right)\left(x^2 + \frac{3}{2}x + \frac{9}{4}\right) = 0$$

; $\boxed{x = -1}$ or $a=1, b=-1, c=1$

$$x - \frac{3}{2} = 0 \quad \text{or} \quad x^2 + \frac{3}{2}x + \frac{9}{4} = 0$$

; $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$$\boxed{x = \frac{3}{2}} \quad \text{or} \quad 4x^2 + 6x + 9 = 0$$

; $x = \frac{-(-1) \pm \sqrt{(-1)^2 - 4(1)(1)}}{2(1)}$

$a=4, b=6, c=9$;

$$x = \frac{1 \pm \sqrt{1-4}}{2} = \frac{1 \pm \sqrt{-3}}{2}$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\boxed{x = \frac{1 \pm \sqrt{3}i}{2}}$$

$$x = \frac{-6 \pm \sqrt{(6)^2 - 4(4)(9)}}{2(4)}$$

$$x = \frac{-6 \pm \sqrt{36(1-4)}}{8}$$

$$x = \frac{-6 \pm \sqrt{36(-3)}}{8}$$

$$x = \frac{-6 \pm 6\sqrt{3}i}{8}$$

$$x = \frac{\cancel{2}(-3 \pm 3\sqrt{3}i)}{\cancel{8}}$$

$$x = \frac{-3 \pm 3\sqrt{3}i}{4}$$

$$\therefore \text{Solution Set} = \left\{ -1, \frac{3}{2}, \frac{-3 \pm 3\sqrt{3}i}{4}, \frac{1 \pm \sqrt{3}i}{2} \right\}$$

Q.5 $x^{\frac{2}{5}} + 8 = 6x^{\frac{1}{5}}$

Solution:

$$x^{\frac{2}{5}} + 8 = 6x^{\frac{1}{5}}$$

$$x^{\frac{2}{5}} - 6x^{\frac{1}{5}} + 8 = 0 \tag{i}$$

$$\text{Put } y = x^{\frac{1}{5}} \Rightarrow y^2 = x^{\frac{2}{5}}$$

Equation (i) becomes

$$y^2 - 6y + 8 = 0$$

$$y^2 - 4y - 2y + 8 = 0$$

$$y(y-4) - 2(y-4) = 0$$

$$(y-4)(y-2) = 0$$

Either

$$y-4=0 \quad \text{or} \quad y-2=0$$

$$y=4 \quad ; \quad y=2$$

$$\text{As } y = x^{\frac{1}{5}} \quad ; \quad \text{As } y = x^{\frac{1}{5}}$$

$$x^{\frac{1}{5}} = 4 \quad ; \quad x^{\frac{1}{5}} = 2$$

$$x = (4)^5 \quad ; \quad x = (2)^5$$

$$x = 1024 \quad ; \quad x = 32$$

$$\therefore \text{Solution Set} = \{32, 1024\}$$

Q.6 $(x+1)(x+2)(x+3)(x+4) = 24$

Solution:

$$(x+1)(x+2)(x+3)(x+4) = 24$$

Re-arranging the equation

$$[(x+1)(x+4)][(x+3)(x+2)] = 24 \quad (\because 1+4=3+2=5)$$

$$(x^2+5x+4)(x^2+5x+6) = 24$$

Put $x^2+5x = y$

Then above equation becomes

$$(y+4)(y+6) = 24$$

$$y^2+10y+24 = 24$$

$$y(y+10) = 0$$

Either

$$y = 0$$

or $y+10=0$

As $x^2+5x = y$

; As $y = x^2+5x$

$$x^2+5x = 0$$

; $x^2+5x+10 = 0$

$$x(x+5) = 0$$

; $a=1, b=5, c=10$

$$\boxed{x=0} \quad \text{or} \quad x+5=0 \quad ; \quad x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$$

$$\boxed{x=-5} \quad x = \frac{-5 \pm \sqrt{(5)^2-4(1)(10)}}{2(1)}$$

$$x = \frac{-5 \pm \sqrt{25-40}}{2}$$

$$\boxed{x = \frac{-5 \pm \sqrt{15}i}{2}}$$

$$\therefore \text{Solution Set} = \left\{ -5, 0, \frac{-5 \pm \sqrt{15}i}{2} \right\}$$

Q.7 $(x-1)(x+5)(x+8)(x+2) - 880 = 0$

Solution:

$$(x-1)(x+5)(x+8)(x+2) - 880 = 0$$

Re-arranging the equation

$$[(x-1)(x+8)][(x+5)(x+2)] = 880 \quad (\because -1+8=5+2=7)$$

$$(x^2 + 7x - 8)(x^2 + 7x + 10) = 880$$

Put $x^2 + 7x = y$

$$(y-8)(y+10) = 880$$

$$y^2 + 2y - 80 - 880 = 0$$

$$y^2 + 2y - 960 = 0$$

$$y^2 + 32y - 30y - 960 = 0$$

$$y(y+32) - 30(y+32) = 0$$

$$(y+32)(y-30) = 0$$

Either

$$y+32=0 \quad \text{or} \quad y-30=0$$

$$\text{As } y = x^2 + 7x \quad ; \quad \text{As } y = x^2 + 7x$$

$$x^2 + 7x + 32 = 0 \quad ; \quad x^2 + 7x - 30 = 0$$

$$a=1, b=7, c=32 \quad ; \quad x^2 + 10x - 3x - 30 = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad ; \quad x(x+10) - 3(x+10) = 0$$

$$x = \frac{-7 \pm \sqrt{(7)^2 - 4(1)(32)}}{2(1)} \quad ; \quad (x+10)(x-3) = 0$$

$$x = \frac{-7 \pm \sqrt{49 - 128}}{2} \quad ; \quad x+10=0 \quad \text{or} \quad x-3=0$$

$$x = \frac{-7 \pm \sqrt{-79}}{2} \quad ; \quad \boxed{x = -10} \quad \text{or} \quad \boxed{x = 3}$$

$$\boxed{x = \frac{-7 \pm \sqrt{79}i}{2}}$$

$$\therefore \text{Solution Set} = \left\{ -10, 3, \frac{-7 \pm \sqrt{79}i}{2} \right\}$$

Q.8 $(x-5)(x-7)(x+6)(x+4) - 504 = 0$

Solution:

$$(x-5)(x-7)(x+6)(x+4) - 504 = 0$$

Re-arranging the equation

$$[(x-5)(x+4)][(x-7)(x+6)] = 504 \quad (\because -5+4 = -7+6 = -1)$$

$$(x^2 - x - 20)(x^2 - x - 42) = 504$$

Put $x^2 - x = y$

$$(y-20)(y-42) = 504$$

$$y^2 - 62y + 840 - 504 = 0$$

$$y^2 - 62y + 336 = 0$$

$$y^2 - 56y - 6y + 336 = 0$$

$$y(y-56) - 6(y-56) = 0$$

$$(y-56)(y-6) = 0$$

Either

$$y-56=0$$

or $y-6=0$

$$y=56$$

; $y=6$

As $x^2 - x = y$

; As $x^2 - x = y$

$$\text{As } x^2 - x - 56 = 0$$

; $x^2 - x - 6 = 0$

$$x^2 - 8x + 7x - 56 = 0$$

; $x^2 - 3x + 2x - 6 = 0$

$$x(x-8) + 7(x-8) = 0$$

; $x(x-3) + 2(x-3) = 0$

$$(x-8)(x+7) = 0$$

; $(x-3)(x+2) = 0$

$$x-8=0 \quad \text{or} \quad x+7=0$$

; $x-3=0 \quad \text{or} \quad x+2=0$

$$\boxed{x=8}$$

;

$$\boxed{x=-7}$$

;

$$\boxed{x=3}$$

;

$$\boxed{x=-2}$$

$$\therefore \text{ solution Set} = \{-7, -2, 3, 8\}$$

Q.9 $(x-1)(x-2)(x-8)(x+5) + 360 = 0$

Solution:

$$(x-1)(x-2)(x-8)(x+5) + 360 = 0$$

$$[(x-1)(x-2)][(x-8)(x+5)] + 360 = 0 \quad (\because -1-2 = -8+5 = -3)$$

$$(x^2 - 3x + 2)(x^2 - 3x - 40) + 360 = 0$$

$$\text{Put } x^2 - 3x = y$$

$$(y + 2)(y - 40) + 360 = 0$$

$$y^2 - 38y - 80 + 360 = 0$$

$$y^2 - 38y + 280 = 0$$

$$y^2 - 28y - 10y + 280 = 0$$

$$y(y - 28) - 10(y - 28) = 0$$

$$(y - 28)(y - 10) = 0$$

Either

$$y - 28 = 0$$

$$\text{or } y - 10 = 0$$

$$\text{As } x^2 - 3x = y$$

$$; \text{ As } x^2 - 3x = y$$

$$x^2 - 3x - 28 = 0$$

$$; \text{ } x^2 - 3x - 10 = 0$$

$$x^2 - 7x + 4x - 28 = 0$$

$$; \text{ } x^2 - 5x + 2x - 10 = 0$$

$$x(x - 7) + 4(x - 7) = 0$$

$$; \text{ } x(x - 5) + 2(x - 5) = 0$$

$$(x - 7)(x + 4) = 0$$

$$; \text{ } (x - 5)(x + 2) = 0$$

$$x - 7 = 0 \quad \text{or} \quad x + 4 = 0$$

$$; \text{ } x - 5 = 0 \quad \text{or} \quad x + 2 = 0$$

$$\boxed{x = 7}$$

;

$$\boxed{x = -4}$$

;

$$\boxed{x = 5}$$

;

$$\boxed{x = -2}$$

$$\therefore \text{ solution Set} = \{-4, -2, 5, 7\}$$

Q.10 $(x + 1)(2x + 3)(2x + 5)(x + 3) = 945$

Solution:

$$(x + 1)(2x + 3)(2x + 5)(x + 3) = 945$$

Re-arranging the equation

$$[(x + 1)(x + 3)][(2x + 3)(2x + 5)] = 945$$

$$\left(\because 1 + 3 = 4, \frac{3}{2} + \frac{5}{2} = \frac{8}{2} = 4 \right)$$

$$(x^2 + 4x + 3)(4x^2 + 10x + 6x + 15) = 945$$

$$(x^2 + 4x + 3)(4x^2 + 16x + 15) = 945$$

$$(x^2 + 4x + 3)(4(x^2 + 4x) + 15) = 945$$

$$\text{Put } x^2 + 4x = y$$

$$(y+3)(4y+15)=945$$

$$4y^2 + 27y + 45 - 945 = 0$$

$$4y^2 + 27y - 900 = 0$$

$$4y^2 + 75y - 48y - 900 = 0$$

$$y(4y+75) - 12(4y+75) = 0$$

$$(y-12)(4y+75) = 0$$

$$y-12=0$$

$$\text{or } 4y+75=0$$

$$\text{As } x^2 + 4x = y$$

$$; \text{ As } x^2 + 4x = y$$

$$x^2 + 4x - 12 = 0$$

$$; 4(x^2 + 4x) + 75 = 0$$

$$x^2 + 6x - 2x - 12 = 0$$

$$; 4x^2 + 16x + 75 = 0$$

$$x(x+6) - 2(x+6) = 0$$

$$; a=4, b=16, c=75$$

$$(x+6)(x-2) = 0$$

$$; x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Either

$$x+6=0 \quad \text{or} \quad x-2=0 \quad ; \quad x = \frac{-16 \pm \sqrt{16^2 - 4(4)(75)}}{2(4)}$$

$$\boxed{x=-6} \quad \text{or} \quad \boxed{x=2} \quad ; \quad x = \frac{-16 \pm \sqrt{256 - 16 \times 75}}{8}$$

$$x = \frac{-16 \pm 4\sqrt{-59}}{8}$$

$$x = \frac{\cancel{4}(-4 \pm \sqrt{59}i)}{\cancel{4}}$$

$$\boxed{x = \frac{-4 \pm \sqrt{59}i}{2}}$$

$$\therefore \text{ Solution Set} = \left\{ -6, 2, \frac{-4 \pm \sqrt{59}i}{2} \right\}$$

Q.11 $(2x-7)(x^2-9)(2x+5)-91=0$

Solution:

$$(2x-7)(x^2-9)(2x+5)-91=0$$

$$(2x-7)(x-3)(x+3)(2x+5)-91=0$$

Re-arranging the equation

$$[(2x-7)(x+3)][(x-3)(2x+5)]-91=0 \quad \left(\because \frac{-7}{2}+3=\frac{5}{2}-3=\frac{-1}{2} \right)$$

$$(2x^2-x-21)(2x^2-x-15)-91=0$$

Put $2x^2-x=y$

$$(y-21)(y-15)-91=0$$

$$y^2-36y+315-91=0$$

$$y^2-36y+224=0$$

$$y^2-28y-8y+224=0$$

$$y(y-28)-8(y-28)=0$$

$$(y-28)(y-8)=0$$

Either

$$y-28=0$$

or $y-8=0$

As $y=2x^2-x$

; As $y=2x^2-x$

$$2x^2-x-28=0$$

; $2x^2-x-8=0$

$$2x^2-8x+7x-28=0$$

; $a=2, b=-1, c=-8$

$$2x(x-4)+7(x-4)=0$$

; $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$$(x-4)(2x+7)=0$$

; $x = \frac{-(-1) \pm \sqrt{(-1)^2 - 4(2)(-8)}}{2(2)}$

$$x-4=0 \quad \text{or} \quad 2x+7=0$$

; $x = \frac{1 \pm \sqrt{1+64}}{4}$

$$\boxed{x=4} \quad ; \quad 2x=-7 \quad ;$$

$$\boxed{x = \frac{1 \pm \sqrt{65}}{4}}$$

; $\boxed{x = \frac{-7}{2}}$

$$\therefore \text{SolutionSet} = \left\{ \frac{-7}{2}, 4, \frac{1 \pm \sqrt{65}}{4} \right\}$$

Q.12 $(x^2 + 6x + 8)(x^2 + 14x + 48) = 105$

Solution:

$$(x^2 + 6x + 8)(x^2 + 14x + 48) = 105$$

$$(x^2 + 4x + 2x + 8)(x^2 + 8x + 6x + 48) = 105$$

$$\{x(x+4) + 2(x+4)\}\{x(x+8) + 6(x+8)\} = 105$$

$$(x+4)(x+2)(x+6)(x+8) = 105$$

Re-arranging the equation

$$[(x+4)(x+6)][(x+2)(x+8)] = 105 \quad (\because 4+6=2+8=10)$$

$$(x^2 + 10x + 24)(x^2 + 10x + 16) = 105$$

Put $x^2 + 10x = y$

$$(y+24)(y+16) = 105$$

$$y^2 + 40y + 384 - 105 = 0$$

$$y^2 + 40y + 279 = 0$$

$$y^2 + 31y + 9y + 279 = 0$$

$$y(y+31) + 9(y+31) = 0$$

$$(y+31)(y+9) = 0$$

Either

$$y+31=0$$

or $y+9=0$

As $y = x^2 + 10x$

; As $y = x^2 + 10x$

$$x^2 + 10x + 31 = 0$$

; $x^2 + 10x + 9 = 0$

$$a=1, b=10, c=31$$

; $x^2 + 9x + x + 9 = 0$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

; $x(x+9) + 1(x+9) = 0$

$$x = \frac{-10 \pm \sqrt{(10)^2 - 4(1)(31)}}{2(1)}$$

; $(x+9)(x+1) = 0$

$$x = \frac{-10 \pm \sqrt{100 - 124}}{2}$$

; $x+9=0$ or $x+1=0$

$$x = \frac{-10 \pm \sqrt{-24}}{2}$$

; $\boxed{x = -9}$; $\boxed{x = -1}$

$$x = \frac{-10 \pm \sqrt{4 \times (-6)}}{2}$$

$$x = \frac{-10 \pm 2\sqrt{6}i}{2}$$

$$x = \frac{-5 \pm \sqrt{6}i}{1}$$

$$x = -5 \pm \sqrt{6}i$$

$$\therefore \text{Solution Set} = \{-9, -1, -5 \pm \sqrt{6}i\}$$

Q.13 $(x^2 + 6x - 27)(x^2 - 2x - 35) = 385$

Solution:

$$(x^2 + 6x - 27)(x^2 - 2x - 35) = 385$$

$$(x^2 + 9x - 3x - 27)(x^2 - 7x + 5x - 35) = 385$$

$$\{x(x+9) - 3(x+9)\}\{x(x-7) + 5(x-7)\} = 385$$

$$(x+9)(x-3)(x-7)(x+5) = 385$$

Re-arranging the equation

$$[(x+9)(x-7)][(x+5)(x-3)] = 385 \quad (\because 9-7=5-3=2)$$

$$(x^2 + 2x - 63)(x^2 + 2x - 15) = 385$$

Put $x^2 + 2x = y$

$$(y - 63)(y - 15) = 385$$

$$y^2 - 78y + 945 - 385 = 0$$

$$y^2 - 78y + 560 = 0$$

$$y^2 - 70y - 8y + 560 = 0$$

$$y(y - 70) - 8(y - 70) = 0$$

$$(y - 70)(y - 8) = 0$$

Either

$$y - 70 = 0 \quad \text{or} \quad y - 8 = 0$$

$$\text{As } x^2 + 2x = y \quad ; \quad \text{As } x^2 + 2x = y$$

$$x^2 + 2x - 70 = 0 \quad ; \quad x^2 + 2x - 8 = 0$$

$$a=1, b=2, c=-70 \quad ; \quad x^2 + 4x - 2x - 8 = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad ; \quad x(x+4) - 2(x+4) = 0$$

$$x = \frac{-2 \pm \sqrt{(2)^2 - 4(1)(-70)}}{2(1)} ; \quad (x+4)(x-2) = 0$$

$$x = \frac{-2 \pm \sqrt{4 + 280}}{2} \quad ; \quad x+4=0 \quad \text{or} \quad x-2=0$$

$$x = \frac{-2 \pm \sqrt{4(1+70)}}{2} \quad ; \quad \boxed{x=-4} \quad ; \quad \boxed{x=2}$$

$$x = \frac{-2 \pm 2\sqrt{71}}{2}$$

$$\boxed{x = -1 \pm \sqrt{71}}$$

$$\therefore \text{Solution Set is } = \{-4, 2, -1 \pm \sqrt{71}\}$$

Q.14 $4 \cdot 2^{2x+1} - 9 \cdot 2^x + 1 = 0$

Solution:

$$4 \cdot 2^{2x+1} - 9 \cdot 2^x + 1 = 0$$

$$4 \cdot 2 \cdot 2^{2x} - 9 \cdot 2^x + 1 = 0$$

$$8 \cdot 2^{2x} - 9 \cdot 2^x + 1 = 0$$

(i)

$$\text{Put } y = 2^x \Rightarrow y^2 = 2^{2x}$$

Equation (i) becomes

$$8y^2 - 9y + 1 = 0$$

$$8y^2 - 8y - y + 1 = 0$$

$$8y(y-1) - 1(y-1) = 0$$

$$(y-1)(8y-1) = 0$$

$$\text{Either } y-1=0 \quad \text{or} \quad 8y-1=0$$

$$\text{As } y = 2^x \quad ; \quad y = 2^x$$

$$2^x - 1 = 0 \quad ; \quad 8 \cdot 2^x - 1 = 0$$

$$2^x = 1 \quad ; \quad 2^x = \frac{1}{8}$$

$$2^x = 2^0 \quad ; \quad 2^x = 2^{-3}$$

$$\boxed{x=0} \quad ; \quad \boxed{x=-3}$$

$$\therefore \text{Solution Set} = \{-3, 0\}$$

Q.15 $2^x + 2^{-x+6} - 20 = 0$

Solution:

$$2^x + 2^{-x+6} - 20 = 0$$

$$2^x + \frac{2^6}{2^x} - 20 = 0$$

$$2^{2x} + 64 - 20 \cdot 2^x = 0 \quad (i)$$

Put $2^x = y \Rightarrow 2^{2x} = y^2$

Equation (i) becomes

$$y^2 - 20y + 64 = 0$$

$$y^2 - 16y - 4y + 64 = 0$$

$$y(y-16) - 4(y-16) = 0$$

$$(y-16)(y-4) = 0$$

Either $y-16=0$ or $y-4=0$

As $y = 2^x$; As $y = 2^x$

$$2^x - 16 = 0 ; 2^x - 4 = 0$$

$$2^x = 16 ; 2^x = 4$$

$$2^x = 2^4 ; 2^x = 2^2$$

$$\boxed{x = 4} ; \boxed{x = 2}$$

$$\therefore \text{Solution Set} = \{2, 4\}$$

Q.16 $4^x - 3 \cdot 2^{x+3} + 128 = 0$

Solution:

$$4^x - 3 \cdot 2^{x+3} + 128 = 0$$

$$(2^2)^x - 3 \cdot 2^3 \cdot 2^x + 128 = 0$$

$$2^{2x} - 24 \cdot 2^x + 128 = 0 \quad (i)$$

Put $y = 2^x \Rightarrow y^2 = 2^{2x}$

Equation (i) becomes

$$y^2 - 24y + 128 = 0$$

$$y^2 - 16y - 8y + 128 = 0$$

$$y(y-16) - 8(y-16) = 0$$

$$(y-16)(y-8) = 0$$

Either $y-16=0$ or $y-8=0$

$y = 16$; $y = 8$

As $y = 2^x$; As $y = 2^x$

$$2^x = 2^4 ; 2^x = 2^3$$

$$\boxed{x = 4} ; \boxed{x = 3}$$

$$\therefore \text{Solution Set} = \{3, 4\}$$

Q.17 $3^{2x-1} - 12 \cdot 3^x + 81 = 0$

Solution:

$$3^{2x-1} - 12 \cdot 3^x + 81 = 0$$

$$\frac{3^{2x}}{3} - 12 \cdot 3^x + 81 = 0$$

$$3^{2x} - 36 \cdot 3^x + 243 = 0 \quad (i)$$

Put $y = 3^x \Rightarrow y^2 = 3^{2x}$

$$y^2 - 36y + 243 = 0$$

$$y^2 - 27y - 9y + 243 = 0$$

$$y(y - 27) - 9(y - 27) = 0$$

$$(y - 27)(y - 9) = 0$$

Either $y - 27 = 0$ or $y - 9 = 0$

As $y = 3^x$; As $y = 3^x$

$3^x = 3^3$; $3^x = 3^2$

$\boxed{x = 3}$; $\boxed{x = 2}$

$\therefore \text{Solution Set} = \{2, 3\}$

Q.18 $\left(x + \frac{1}{x}\right)^2 - 3\left(x + \frac{1}{x}\right) - 4 = 0$

Solution:

$$\left(x + \frac{1}{x}\right)^2 - 3\left(x + \frac{1}{x}\right) - 4 = 0 \quad (i)$$

Put $x + \frac{1}{x} = y$

Equation (i) becomes

$$y^2 - 3y - 4 = 0$$

$$y^2 - 4y + y - 4 = 0$$

$$y(y - 4) + 1(y - 4) = 0$$

$$(y - 4)(y + 1) = 0$$

Either

$$y - 4 = 0$$

or $y + 1 = 0$

$$\text{As } y = x + \frac{1}{x}$$

$$; \quad \text{As } y = x + \frac{1}{x}$$

$$x + \frac{1}{x} - 4 = 0$$

$$; \quad x + \frac{1}{x} + 1 = 0$$

$$x^2 - 4x + 1 = 0$$

$$; \quad x^2 + x + 1 = 0$$

$$a = 1, b = -4, c = 1$$

$$; \quad a = 1, b = 1, c = 1$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$; \quad x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-(-4) \pm \sqrt{(-4)^2 - 4(1)(1)}}{2(1)}$$

$$; \quad x = \frac{-1 \pm \sqrt{(1)^2 - 4(1)(1)}}{2(1)}$$

$$x = \frac{4 \pm \sqrt{12}}{2}$$

$$; \quad x = \frac{-1 \pm \sqrt{-3}}{2}$$

$$x = \frac{4 \pm 2\sqrt{3}}{2}$$

$$; \quad \boxed{x = \frac{-1 \pm \sqrt{3}i}{2}}$$

$$\boxed{x = 2 \pm \sqrt{3}}$$

$$\therefore \text{Solution Set} = \left\{ 2 \pm \sqrt{3}, \frac{-1 \pm \sqrt{3}i}{2} \right\}$$

Q.19 $x^2 + x - 4 + \frac{1}{x} + \frac{1}{x^2} = 0$

Solution:

$$x^2 + x - 4 + \frac{1}{x} + \frac{1}{x^2} = 0$$

$$x^2 + \frac{1}{x^2} + x + \frac{1}{x} - 4 = 0$$

$$x^2 + \frac{1}{x^2} + 2 + x + \frac{1}{x} - 4 - 2 = 0$$

$$\left(x + \frac{1}{x}\right)^2 + \left(x + \frac{1}{x}\right) - 6 = 0$$

(i)

$$\text{Put } y = x + \frac{1}{x} \Rightarrow y^2 = \left(x + \frac{1}{x}\right)^2$$

Equation (i) becomes

$$y^2 + y - 6 = 0$$

$$y^2 + 3y - 2y - 6 = 0$$

$$y(y+3) - 2(y+3) = 0$$

$$(y+3)(y-2) = 0$$

Either

$$y+3=0$$

or $y-2=0$

As $y = x + \frac{1}{x}$; As $y = x + \frac{1}{x}$

$x + \frac{1}{x} + 3 = 0$; $x + \frac{1}{x} - 2 = 0$

$x^2 + 3x + 1 = 0$; $x^2 - 2x + 1 = 0$

$a=1, b=3, c=1$; $(x-1)^2 = 0$

$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$; $x - 1 = 0$

$x = \frac{-3 \pm \sqrt{(3)^2 - 4(1)(1)}}{2(1)}$; $\boxed{x=1}$

$x = \frac{-3 \pm \sqrt{9-4}}{2}$

$\boxed{x = \frac{-3 \pm \sqrt{5}}{2}}$

$\therefore \text{Solution Set} = \left\{ 1, \frac{-3 \pm \sqrt{5}}{2} \right\}$

Q.20 $\left(x - \frac{1}{x}\right)^2 + 3\left(x + \frac{1}{x}\right) = 0$

Solution:

$$\left(x - \frac{1}{x}\right)^2 + 3\left(x + \frac{1}{x}\right) = 0$$

$$x^2 + \frac{1}{x^2} - 2 + 3\left(x + \frac{1}{x}\right) = 0$$

$$x^2 + \frac{1}{x^2} + 2 + 3\left(x + \frac{1}{x}\right) - 2 - 2 = 0$$

$$\left(x + \frac{1}{x}\right)^2 + 3\left(x + \frac{1}{x}\right) - 4 = 0 \quad (i)$$

$$\text{Put } x + \frac{1}{x} = y \Rightarrow \left(x + \frac{1}{x}\right)^2 = y^2$$

Equation (i) becomes

$$y^2 + 3y - 4 = 0$$

$$y^2 + 4y - y - 4 = 0$$

$$y(y+4) - 1(y+4) = 0$$

$$(y+4)(y-1) = 0$$

Either

$$y+4=0$$

$$\text{or } y-1=0$$

$$\text{As } y = x + \frac{1}{x} \quad ; \quad \text{As } y = x + \frac{1}{x}$$

$$x + \frac{1}{x} + 4 = 0 \quad ; \quad x + \frac{1}{x} - 1 = 0$$

$$x^2 + 4x + 1 = 0 \quad ; \quad x^2 - x + 1 = 0$$

$$a=1, b=4, c=1 \quad ; \quad a=1, b=-1, c=1$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad ; \quad x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-4 \pm \sqrt{(4)^2 - 4(1)(1)}}{2(1)} \quad ; \quad x = \frac{-(-1) \pm \sqrt{(-1)^2 - 4(1)(1)}}{2(1)}$$

$$x = \frac{-4 \pm \sqrt{12}}{2} \quad ; \quad x = \frac{1 \pm \sqrt{-3}}{2}$$

$$x = \frac{-4 \pm 2\sqrt{3}}{2} \quad ; \quad x = \frac{1 \pm \sqrt{3}i}{2}$$

$$\boxed{x = -2 \pm \sqrt{3}} \quad ; \quad \boxed{x = \frac{1 \pm \sqrt{3}i}{2}}$$

$$\therefore \text{Solution Set} = \left\{ -2 \pm \sqrt{3}, \frac{1 \pm \sqrt{3}i}{2} \right\}$$

Q.21 $2x^4 - 3x^3 - x^2 - 3x + 2 = 0$

Solution:

$$2x^4 - 3x^3 - x^2 - 3x + 2 = 0$$

Dividing by ' x^2 ,

$$2x^2 - 3x - 1 - \frac{3}{x} + \frac{2}{x^2} = 0$$

$$2\left(x^2 + \frac{1}{x^2}\right) - 3\left(x + \frac{1}{x}\right) - 1 = 0$$

$$2\left(x^2 + \frac{1}{x^2} + 2\right) - 3\left(x + \frac{1}{x}\right) - 1 - 4 = 0$$

$$2\left(x + \frac{1}{x}\right)^2 - 3\left(x + \frac{1}{x}\right) - 5 = 0 \quad (i)$$

$$\text{Put } y = x + \frac{1}{x} \Rightarrow y^2 = \left(x + \frac{1}{x}\right)^2$$

Equation (i) becomes

$$2y^2 - 3y - 5 = 0$$

$$2y^2 - 5y + 2y - 5 = 0$$

$$y(2y - 5) + 1(2y - 5) = 0$$

$$(2y - 5)(y + 1) = 0$$

Either

$$2y - 5 = 0$$

$$\text{or } y + 1 = 0$$

$$\text{As } y = x + \frac{1}{x} \quad ; \quad \text{As } y = x + \frac{1}{x}$$

$$2\left(x + \frac{1}{x}\right) - 5 = 0 \quad ; \quad x + \frac{1}{x} + 1 = 0$$

$$2x + \frac{2}{x} - 5 = 0 \quad ; \quad x^2 + x + 1 = 0$$

$$2x^2 - 5x + 2 = 0 \quad ; \quad a = 1, b = 1, c = 1$$

$$2x^2 - 4x - x + 2 = 0 \quad ; \quad x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$2x(x - 2) - 1(x - 2) = 0 \quad ; \quad x = \frac{-1 \pm \sqrt{(1)^2 - 4(1)(1)}}{2(1)}$$

$$(x - 2)(2x - 1) = 0 \quad ; \quad x = \frac{-1 \pm \sqrt{-3}}{2}$$

$$x - 2 = 0, 2x - 1 = 0 \quad ; \quad x = \frac{-1 \pm \sqrt{3}i}{2}$$

$$x = 2 \quad x = \frac{1}{2}$$

$$\therefore \text{ Solution Set} = \left\{ \frac{1}{2}, 2, \frac{-1 \pm \sqrt{3}i}{2} \right\}$$

Q.22 $2x^4 + 3x^3 - 4x^2 - 3x + 2 = 0$

Solution:

$$2x^4 + 3x^3 - 4x^2 - 3x + 2 = 0$$

Dividing by ' x^2 '

$$2x^2 + 3x - 4 - \frac{3}{x} + \frac{2}{x^2} = 0$$

$$2\left(x^2 + \frac{1}{x^2}\right) + 3\left(x - \frac{1}{x}\right) - 4 = 0$$

$$2\left(x^2 + \frac{1}{x^2} - 2\right) + 3\left(x - \frac{1}{x}\right) = 0$$

$$2\left(x - \frac{1}{x}\right)^2 + 3\left(x - \frac{1}{x}\right) = 0 \quad (i)$$

Put $x - \frac{1}{x} = y \Rightarrow \left(x - \frac{1}{x}\right)^2 = y^2$

Equation (i) becomes

$$2y^2 + 3y = 0$$

$$y(2y + 3) = 0$$

Either

$$y = 0$$

$$\text{As } x - \frac{1}{x} = y$$

$$x - \frac{1}{x} = 0$$

$$x^2 - 1 = 0$$

$$x^2 = 1$$

$$x = \pm 1$$

$$\text{or } 2y + 3 = 0$$

$$; \text{ As } y = x - \frac{1}{x}$$

$$; 2\left(x - \frac{1}{x}\right) + 3 = 0$$

$$; 2x - \frac{2}{x} + 3 = 0$$

$$; 2x^2 + 3x - 2 = 0$$

$$; 2x^2 + 4x - x - 2 = 0$$

$$; 2x(x + 2) - 1(x + 2) = 0$$

$$; (x + 2)(2x - 1) = 0$$

$$; x + 2 = 0 \quad \text{or} \quad 2x - 1 = 0$$

$$; \boxed{x = -2} \quad ; \quad \boxed{x = \frac{1}{2}}$$

$$\therefore \text{ Solution Set} = \left\{-1, 1, -2, \frac{1}{2}\right\}$$

Q.23 $6x^4 - 35x^3 + 62x^2 - 35x + 6 = 0$

Solution:

$$6x^4 - 35x^3 + 62x^2 - 35x + 6 = 0$$

Dividing by ' x^2 '

$$6x^2 - 35x + 62 - \frac{35}{x} + \frac{6}{x^2} = 0$$

$$6\left(x^2 + \frac{1}{x^2}\right) - 35\left(x + \frac{1}{x}\right) + 62 = 0$$

$$6\left(x^2 + \frac{1}{x^2} + 2\right) - 35\left(x + \frac{1}{x}\right) + 62 - 12 = 0$$

$$6\left(x + \frac{1}{x}\right)^2 - 35\left(x + \frac{1}{x}\right) + 50 = 0 \quad (i)$$

Put $x + \frac{1}{x} = y \Rightarrow \left(x + \frac{1}{x}\right)^2 = y^2$

Equation (i) becomes

$$6y^2 - 35y + 50 = 0$$

$$6y^2 - 20y - 15y + 50 = 0$$

$$2y(3y - 10) - 5(3y - 10) = 0$$

$$(3y - 10)(2y - 5) = 0$$

Either

$$3y - 10 = 0$$

or $2y - 5 = 0$

As $x + \frac{1}{x} = y$

; As $x + \frac{1}{x} = y$

$$3\left(x + \frac{1}{x}\right) - 10 = 0$$

; $2\left(x + \frac{1}{x}\right) - 5 = 0$

$$3x + \frac{3}{x} - 10 = 0$$

; $2x + \frac{2}{x} - 5 = 0$

$$3x^2 + 3 - 10x = 0$$

; $2x^2 - 5x + 2 = 0$

$$3x^2 - 10x + 3 = 0$$

; $2x^2 - 4x - x + 2 = 0$

$$3x^2 - 9x - x + 3 = 0$$

; $2x(x - 2) - 1(x - 2) = 0$

$$3x(x - 3) - 1(x - 3) = 0$$

; $(x - 2)(2x - 1) = 0$

$$(x - 3)(3x - 1) = 0$$

$x - 3 = 0$ or $3x - 1 = 0$; $x - 2 = 0$ or $2x - 1 = 0$

$$\boxed{x = 3}$$

$$\boxed{x = \frac{1}{3}}$$

; $\boxed{x = 2}$

$$\boxed{x = \frac{1}{2}}$$

$$\therefore \text{Solution Set} = \left\{\frac{1}{3}, \frac{1}{2}, 2, 3\right\}$$

Q.24 $x^4 - 6x^2 + 10 - \frac{6}{x^2} + \frac{1}{x^4} = 0$

Solution:

$$x^4 - 6x^2 + 10 - \frac{6}{x^2} + \frac{1}{x^4} = 0$$

$$x^4 + \frac{1}{x^4} - 6\left(x^2 + \frac{1}{x^2}\right) + 10 = 0$$

$$x^4 + \frac{1}{x^4} + 2 - 6\left(x^2 + \frac{1}{x^2}\right) + 10 - 2 = 0$$

$$\left(x^2 + \frac{1}{x^2}\right)^2 - 6\left(x^2 + \frac{1}{x^2}\right) + 8 = 0 \quad (i)$$

$$\text{Put } x^2 + \frac{1}{x^2} = y \Rightarrow \left(x^2 + \frac{1}{x^2}\right)^2 = y^2$$

Equation (i) becomes

$$y^2 - 6y + 8 = 0$$

$$y^2 - 4y - 2y + 8 = 0$$

$$y(y - 4) - 2(y - 4) = 0$$

$$(y - 4)(y - 2) = 0$$

Either

$$y - 4 = 0 \quad \text{or} \quad y - 2 = 0$$

$$\text{As } y = x^2 + \frac{1}{x^2} \quad ; \quad y = x^2 + \frac{1}{x^2}$$

$$x^2 + \frac{1}{x^2} - 4 = 0 \quad ; \quad x^2 + \frac{1}{x^2} - 2 = 0$$

$$x^4 - 4x^2 + 1 = 0 \quad ; \quad x^4 - 2x^2 + 1 = 0$$

$$\text{Put } x^2 = t \Rightarrow x^4 = t^2 \quad ; \quad \text{Put } x^2 = t \Rightarrow x^4 = t^2$$

$$t^2 - 4t + 1 = 0 \quad ; \quad t^2 - 2t + 1 = 0$$

$$a = 1, b = -4, c = 1 \quad ; \quad (t - 1)^2 = 0$$

$$t = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad ; \quad t - 1 = 0$$

$$t = \frac{-(-4) \pm \sqrt{(-4)^2 - 4(1)(1)}}{2(1)} \quad ; \quad t = 1$$

$$t = \frac{4 \pm \sqrt{16 - 4}}{2} = \frac{4 \pm \sqrt{12}}{2} \quad ; \quad \text{As } t = x^2$$

$$t = \frac{4 \pm 2\sqrt{3}}{2} \quad ; \quad x^2 = 1$$

$$t = 2 \pm \sqrt{3}$$

$$\text{As } t = x^2 \quad ; \quad \boxed{x = \pm 1}$$

$$x^2 = 2 \pm \sqrt{3}$$

$$\boxed{x = \pm \sqrt{2 \pm \sqrt{3}}}$$

$$\therefore \text{ Solution Set} = \left\{ \pm 1, \pm \sqrt{2 \pm \sqrt{3}} \right\}$$